Energy Reports 8 (2022) 24–36

Contents lists available at ScienceDirect

Energy Reports

journal homepage: www.elsevier.com/locate/egyr

Research paper

Electrical efficiency of the photovoltaic/thermal collectors cooled by nanofluids: Machine learning simulation and optimization by evolutionary algorithm

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ARTICLE INFO

Article history:
Received 22 August 2021
Received in revised form 14 November 2021
Accepted 22 November 2021
Available online 10 December 2021

Keywords:
Photovoltaic/thermal collector
Nanofluids
Electrical efficiency enhancement
Machine learning

ABSTRACT

Photovoltaic/thermal (PV/T) are high-tech devices to transform solar radiation into electrical and thermal energies. Nano-coolants are recently considered to enhance the efficiency of PV/T systems. There is no accurate model to predict/optimize the PV/T systems' electrical efficiency cooled by nano-coolants. Therefore, this research employs machine-learning approaches to simulate PV/T system electrical performance cooled by water-based nanofluids. The best topology of artificial neural networks, least-squares support vector regression, and adaptive neuro-fuzzy inference systems (ANFIS) are found by trial-and-error and statistical analyses. The ANFIS is found as the best method for simulation of the electrical performance of the considered solar system. This approach predicted 200 experimental datasets with the absolute average relative deviation (AARD) of 13.6%, mean squared error (MSE) of 2.548, and $R^2 = 0.9534$. Furthermore, the ANFIS model predicts a new external database containing 63 samples with the AARD=15.21%. The optimization stage confirms that 30 lit/hr of water-silica nano-coolant (3wt%, 12.5 nm) at radiation intensity of 788.285 W/m^2 is the condition that maximizes electrical efficiency. In this optimum condition, the enhancement in the PV/T electrical efficiency is 27.7%. Finally, the fabricated ANFIS model has been utilized for generating several pure simulation predictions that have never been published before.

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1. Introduction

Underground hydrocarbon reservoirs are the leading resources for satisfying the demand for energy in recent decades (Vaferi and Eslamloueyan, 2015). The emission of produced greenhouse gases by human activities is believed to cause climate change and global warming (Karimi et al., 2021b; Rahimpour et al., 2011b,a). Replacing fossil fuels with green (Nouriani and Moradi, 2020; Rouhi et al., 2020) and renewable energy sources (Eshra and Salem, 2020; Nouriani and Moradi, 2021) can decrease air pollution. Photovoltaic (PV) modules are currently leading technologies to benefit from solar radiation (Syahputra and Soesanti, 2021). The semiconductor part of the PV module is responsible for directly converting the sun's shortwave radiations to electricity (Akel et al., 2021). Heat is also generated due to the low energy waves absorbed by the PV cells. According to the general equation of diodes, the low yield of PV cells (10%–30%) is related to a rise in the panel temperature during heat generation (Florschuetz, 1979).

Enhancing the performance of such systems is one of the main under research topics (Zainal et al., 2019). Coupling the PV module with a heat recovery system increases the electrical efficiency and thermal energy (Ahmad et al., 2018; Moradi et al., 2013). In the past few years, photovoltaic/thermal (PV/T) modules have been an attractive research field, and their performances were assessed by carrying out many experiments (Kidegho et al., 2021). One of the most exciting design aspects of the PV/T systems is the characterization of the working fluid in their heat recovery unit (Aberoumand et al., 2018; Al-Waeli et al., 2017; Ebaid et al., 2018; Sardarabadi et al., 2017, 2014). The system performance was examined to determine the effect of ambient temperature, solar radiation, chamber of condensation, and wind speed, and 57% total efficiency was obtained from the experiments (Moh'd A and Al-Ammari, 2016).

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https://doi.org/10.1016/j.egyr.2021.11.252
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Nano-sized solid materials have found great popularity in diverse fields of engineering applications (Charchi et al., 2020; Mokhtari et al., 2017). Modifying the conventional coolant properties by adding nanoparticles for potential applications in the cooling circuit of PV/T systems has been the main target of many experimental investigations (Aberoumand et al., 2018; Al-Waeli et al., 2017; Ebaid et al., 2018; Sardarabadi et al., 2017, 2014). Nanofluids' density, specific heat capacity, dynamic viscosity, and thermal conductivity are essential thermophysical properties influencing the solar collector's thermal and electrical efficiency (Al-Waeli et al., 2017; Ebaid et al., 2018; Sardarabadi et al., 2017). Therefore, researchers have recently focused on replacing the conventional working fluids in solar exchangers with nanofluids (Ouyang et al., 2021; Yahya et al., 2021). Since even the low concentration of nanoparticles (Kaabipour and Hemmati, 2021; Seaberg et al., 2020) in conventional fluids significantly increases their thermal properties (Ebaid et al., 2018), the nanofluids are considered novel cooling/heating fluids (Askari et al., 2021). Luo et al. applied synthesized nanofluids (SiO_2, graphite, TiO_2, Ag, Al_2O_3, Cu, and CNT nanoparticles in the Texatherm oil base fluid) in the direct absorption solar collectors (Luo et al., 2014). Both efficiency and output temperature of the solar collector is enhanced by utilizing nanofluid. Michael and Iniyan used a CuO/water nanofluid to increase the efficiency of a solar water heater and a solar PV module and eliminate their inherent drawbacks (Michael and Iniyan, 2015). Ghadiri et al. observed a substantial enhancement of about 45% in total efficiency in indoor experiments using a solar simulator with constant solar radiations of 600 W/m^2 and 1100 W/m^2 (Ghadiri et al., 2015).

In recent years researchers have benefitted from artificial intelligence (AI) techniques in many different fields (Khandakar et al., 2021a, 2019, 2021b). These AI-based methods have the advantage of fast finding a proper relation between the inputs and outputs (Amar et al., 2020a; Azizi and Pleimling, 2021; Çolak, 2021; Wang et al., 2020). Another study implemented the recurrent neural network and ant lion optimizer to predict the power obtained from PV systems (Roy et al., 2019). The solar irradiation and panel temperature are independent variables, while the generated energy was the target variable. The training step is achieved by using different machine learning techniques. Some researchers observed increasing pressure drop and pumping power by adding nanoparticles to base fluids (Routbort et al., 2011). However, increasing the pumping power is compensated by reducing the required flow rate for the same amount of energy transfer (Routbort et al., 2011).

Despite many experimental studies on the PV/T systems, a few models are available to predict their efficiency. The available correlations need precise information about thermophysical properties of base fluids, nanoparticles, and nanofluids that are often accessible by expensive and time-consuming experimentations. Furthermore, they not only present high levels of uncertainty (Joshi et al., 2009; Sarhaddi et al., 2010; Slimani et al., 2016), the nanofluid coolants even increase their uncertainty. The mean square error (MSE) values of 166.41 (Slimani et al., 2016), 268.3 (Joshi et al., 2009), and 158.6 (Sarhaddi et al., 2010) are reported for predicting the short-circuit current in the PV/T collector. Moreover, the effects of nanofluid properties and operating conditions on the electrical efficiency of the PV/T systems have not been analyzed deeply. Therefore, this paper explores different AI-based models to find the most accurate one for predicting the PV/T system electrical efficiency cooled by the nanofluids. The ANFIS (adaptive network-based fuzzy inference systems), MLP (multi-layer perceptron), CFF (cascade feedforward), RBF (radial basis function), GR (generalized regression) neural networks, and LS-SVR (least-squares support vector regression) are employed to simulate the PV/T system electrical efficiency. The trial-and-error scenario and statistical investigations approves the ANFIS with the subtractive clustering membership function trained by the hybrid algorithm shows the most accurate predictions for the given task. Finally, the differential evolution (DE) algorithm is employed to determine the condition that maximizes the PV/T electrical efficiency. To the best of our knowledge, there have been conducted no study for maximizing the electrical performance of the PV/T systems cooled by different nanofluids.

2. Materials and methods

This section reviews the machine learning methods utilized in the current study for simulating the electrical efficiency of the photovoltaic/thermal collectors. Furthermore, some details about the photovoltaic module and available experimental data in the literature are presented in this section.

2.1. Soft computing approaches

This section extensively explains the working procedure of the three considered machine learning classes for estimating the electrical efficiency of the photovoltaic/thermal collectors, namely, artificial neural networks (MLP, CFF, GR, RBF), ANFIS, and LS-SVR.

2.1.1. Artificial neural networks

The MLP (Esmaeili-Faraj et al., 2021; Vaferi et al., 2016), CFF (Amar, 2020), GR (Mahmoodi et al., 2018), RBF (Syah et al., 2021), and Elman (Vaferi et al., 2015) are among the most widely used types of artificial neural networks. All these intelligent networks constitute artificial neurons (Asgharpour et al., 2020), i.e., the smallest meaningful part of the ANN. Eq. (1) expresses the mathematical operations performed inside the artificial neuron. This equation needs neuron's entry values (x), weight (w), bias (θ), and an activation function (ϕ) to calculate the neuron's output signal (Karimi et al., 2018).

$$Neuron\,output = \phi\ (net) = \phi\left(\sum_{i=1}^{Z} w_i x_i + \theta\right) \tag{1}$$

Different ANN types are possibly be constructed by placing artificial neurons in predefined architectures. The ANN models' structure is often defined by the numbers of their neuronic layers and neurons in each layer (Amidi et al., 2021). The ANN models may have arbitrary numbers of neuronic layers (Yousefi et al., 2020), but the literature approves that two layers (hidden and output) are enough for major modeling tasks (Cybenko, 1989). Therefore, all ANN models in the current study have been developed with only two neuronic layers.

The main difference among MLP, CFF, GR, and RBF neural networks is associated with their activation functions. The MLP and CFF neural networks in this study have tangent (2)(a) and logarithm (2)(b) sigmoid activation functions in their hidden and output layers, respectively. On the other hand, the GR and RBF neural networks have Gaussian (2)(c) and linear (2)(d) activation functions in their neuronic layers. The s in Eq. (2)(a) shows the spread factor of the Gaussian activation function.

$$\phi\ (net) = \begin{cases} (2/\ (1 + \exp\ (-2 \times net))) - 1 & (a) \\ 1/\ (1 + \exp\ (-net)) & (b) \\ \exp\left(-net^2/2s^2\right) & (c) \\ net & (d) \end{cases} \tag{2}$$

Moreover, in the MLP, RBF, and GR neural networks, available neurons in each layer are only allowed to connect to the next layer. On the other hand, the neurons in each layer of the CFF

neural networks are connected to neurons in all subsequent layers.

ANN type, structure, and independent and dependent variables are known in the training stage, and only weights and biases are unknown. The training algorithm is then applied to adjust the hyper-parameters of the ANN model. This study uses the Levenberg–Marquardt optimization algorithm (Ghanbari and Vaferi, 2015; Karimi et al., 2020) to train the MLP and CFF neural networks. The GR and RBF are one-pass trainable neural networks, and they need no optimization algorithm in the training stage (Hagan et al., 1997).

2.1.2. Adaptive neuro-fuzzy inference systems

A particular name (i.e., ANFIS) has been assigned to the computational approach that combines the ANN and fuzzy logic (Karimi et al., 2021a). This combination allows obtaining accurate solutions even for highly non-linear systems (Jiang et al., 2021). Fuzzy logic improved the learning ability of an ANN (Heydarian and Jolai, 2018). Depending on the structures, different fuzzy inference systems (FIS), such as Mamdani and Takagi–Sugeno, may be incorporated in the ANFIS architecture. The first FIS-type includes logical definitions in the development of fuzzy if-then rules. On the other hand, the other FIS-type creates the if-then rules by implementing accessible experimental data. The Takagi–Sugeno inference system represents the nonlinear dependency of the input and output variable in the ANFIS method. For an ordinary ANFIS structure that contains two input parameters (i.e., Z_1 and Z_2), *if-then* rules are expressed as follows:

$$\text{If } Z_1 \text{ is } B_1 \text{ and } Z_2 \text{ is } A_1 \text{ then } g_1 = m_1 Z_1 + n_1 Z_2 + r_1 \tag{3}$$

$$\text{If } Z_1 \text{ is } B_2 \text{ and } Z_2 \text{ is } A_2 \text{ then } g_2 = m_2 Z_1 + n_2 Z_2 + r_2 \tag{4}$$

$$\text{If } Z_1 \text{ is } B_1 \text{ and } Z_2 \text{ is } A_2 \text{ then } g_3 = m_3 Z_1 + n_3 Z_2 + r_3 \tag{5}$$

$$\text{If } Z_1 \text{ is } B_2 \text{ and } Z_2 \text{ is } A_1 \text{ then } g_4 = m_4 Z_1 + n_4 Z_2 + r_4 \tag{6}$$

In these equations, Bi and Ai (i = 1,2) are fuzzy sets for Z_1 and Z_2, respectively. Moreover, g shows the output signal.

To determine the nonlinear correlation between input/output parameters, a standard ANFIS can be used. There are five layers in the ANFIS structure (i.e., fuzzification, fuzzy rule, normalization, defuzzification, and output). The first layer involves the application of input membership functions on input parameters for grading input ranges. The second layer consists of the assignment of weights to the range of input data. The third layer involves the standardization of weighted input data. The fourth layer involves passing the rules defined by shifting the weighted input data herein. The rule outputs are collected and combined and then transformed into desired output values by defuzzification. ANFIS standardizes the weighted input data through self-training. The ANFIS training is often accomplished by a hybrid and back-propagation method.

2.1.3. Least-squares support vector regression

The LS-SVR is a supervised training AI-based model applicable for both regression (Amar et al., 2020b; Daryayehsalameh et al., 2021) and classification (Moosavi et al., 2021) problems. The LS-SVR incorporates a kernel function (φ) in Eq. (7) to transfer independent variables to a multi-dimensional space (Nabavi et al., 2021).

$$g(z) = w^T(z)\varphi(z) + b \tag{7}$$

where z shows the independent variables, w^T is the transpose of the weight matrix, and b is the bias. Linear (Eq. (8)a), polynomial (Eq. (8)b), and Gaussian (Eq. (8)c) are the main kernel functions in the field of LS-SVR methodology (Nabavi et al., 2021). In Eq. (8), T indicates transpose operation, t, d, and σ are inherent characteristics of the kernel functions.

$$\varphi\left(z_i,\, z_j\right) = \begin{cases} {z_i}^T z_j & (a) \\ \left({z_i}^T z_j + t\right)^d & (b) \\ \exp\left(-\left\|z_i - z_j\right\|^2 / 2\sigma^2\right) & (c) \end{cases} \tag{8}$$

The cost function represented by Eq. (9) helps to adjust w and b parameters of the LS-SVR model during the training stage (Nabavi et al., 2021):

$$\text{Cost function} = \frac{1}{2} w^T + c \sum_{k=1}^{N} (\xi_k - {\xi^*}_k) \tag{9}$$

where ε is the precision of the function estimations, c is the tuning factor which determines the deviation from ε, and slack variables (ξk and ξk$*$) are the allowable margin of error.

2.2. Photovoltaic/thermal collector

Generally, solar energy modules are divided into two categories: Some devices convert solar radiation into thermal energy, and others generate electrical energy from solar irradiation (Goetzberger and Hoffmann, 2005). PV panels at best conditions convert between 6 and 20% of the received radiation into electrical energy, and the rest of the radiation raises the panel temperature (Goetzberger and Hoffmann, 2005). This increase in the panel temperature reduces the collector's efficiency so that in crystalline mono panels, with each degree of temperature increase of 0.2–0.45% decrease in panel efficiency occurs (Thopil et al., 2020). Therefore, according to the above description, PV efficiency can be increased by cooling down the panel surface, and the resulting heat can be used in other applications. To cool the panel, passing the fluid under the solar panel, heat transfer between the panel and the fluid causes the panel temperature to drop and the fluid to heat up.

The PV/T that generates thermal and electrical energies from solar radiation combines the solar collector and photovoltaic panel. PV/T collectors have the following advantages over conventional solar collectors:

1- Using the optimal installation space due to the combination of two systems in a single system

2- Increasing the efficiency of the photovoltaic module by reducing its temperature using the operating fluid circulating in the solar collector

3- No need for an external electric source to circulate the operating fluid in the solar collector using the electricity generated in the photovoltaic module

4- Increasing energy efficiency compared to conventional systems

The PV/T collector's performance depends on air temperature, solar intensity, wind speed, photovoltaic module surface properties, entry/effluent fluid temperatures, coolant flow rate, open-circuit voltage, and collector dimensions. The electrical efficiency (η) of the PV/T module is calculated as follows (Al-Waeli et al., 2017):

$$\eta = \frac{P}{A \times I} \tag{10}$$

where P is the electrical power generated by the panel, A is the PV/T area, and I shows the solar radiation intensity.

2.3. Literature experimental dataset

All of the machine learning approaches need to be trained by experimental data about the considered problem. Hence, this

Table 1
Summary of experimental datasets (independent and dependent variables) and their available ranges.

Ranges of independent variables					Ranges of dependent variable	Reference
Nanoparticle	Size (nm)	Radiation intensity (W/m^2)	Nanoparticle concentration (wt%)	Coolant flow rate (l/h)	Electrical efficiency increase (%)	
Ag	20–50	422.4–864.2	2–5	9-417.6	1.51–16.36	Aberoumand et al. (2018), Khanjari et al. (2016)
Al_2O_3	45–50	78–1000	0.3–5	4.9–540	0.14–99.82	Al-Waeli et al. (2017), Khanjari et al. (2017, 2016) and Sangeetha et al. (2021)
CuO	40	700–1000	4	30	9.79–13.5	Al-Waeli et al. (2017)
SiC	55	700–1000	4	30	18.35–24.54	Al-Waeli et al. (2017)
SiO_2	12.5	622.6–1089.5	1–3	30	4.35–27.7	Sardarabadi et al. (2014)
TiO_2	15–45	78-760.7	0.1–0.3	30–540	0.69–99.83	Ebaid et al. (2018) and Sangeetha et al. (2021)
ZnO	40	627.6–1062.9	0.2–6	30–50	0.61–91.12	Hosseinzadeh et al. (2018) and Sardarabadi et al. (2017)

Table 2
Available thermophysical properties for nanoparticle/nanofluid in the source articles.

Thermal conductivity	Dynamic viscosity	Density	Specific heat capacity	
×	×	×	×	Aberoumand et al. (2018)
Nanofluid	Nanofluid	Nanofluid	×	Ebaid et al. (2018)
×	×	×	×	Sardarabadi et al. (2014)
Nanoparticles	×	Nanoparticles	Nanoparticles	Sardarabadi et al. (2017)
Nanoparticles, Nanofluid	Nanofluid	Nanoparticles, Nanofluid	×	Al-Waeli et al. (2017)

study utilizes 263 experimental datasets for electrical efficiency enhancement of the photovoltaic/thermal system reported in nine different research articles (Aberoumand et al., 2018; Al-Waeli et al., 2017; Ebaid et al., 2018; Hosseinzadeh et al., 2018; Khanjari et al., 2017, 2016; Sangeetha et al., 2021; Sardarabadi et al., 2017, 2014).

Table 1 reports that the gathered databank comprises independent (nanofluid flow rate, radiation intensity, type and size of nanoparticles and their concentration in water) and dependent (electrical efficiency enhancement) variables of the PV/T solar collector. It should be mentioned that this enhancement is defined as the ratio of PV/T electrical efficiency obtained by nanofluids and water as a coolant minus one. This table shows that our experimental database includes seven different nanoparticles and relatively wide ranges for the operating conditions.

As mentioned earlier, the current study aims to simulate the electrical efficiency of the photovoltaic/thermal collectors cooled by different nanofluids. Since the electrical efficiency of the PV/T collector depends on the thermophysical properties (specific heat capacity, dynamic viscosity, density, and thermal conductivity) of the nanofluids, it is better to mention the reported ones in the source articles. Table 2 reports the available thermophysical properties for either nanoparticle and nanofluid in the source articles.

3. Results and discussions

As mentioned earlier, the present work aims to propose a reliable AI-based model for estimating the PV/T system's electrical efficiency that cools down by nanofluids. We focused on ANN, LS-SVR, and ANFIS as three leading soft computing approaches. This section explains the procedure of finding the best ANN, ANFIS, and LS-SVR models first. After that, the best machine learning (ML) technique is found by comparing the predictive performances of the best ANN, ANFIS, and LS-SVR models. Finally, the best ML approach is employed to evaluate the effect of nanofluids' properties and operating and environmental conditions on enhancing the photovoltaic/thermal electrical efficiency.

3.1. The best structural features of the machine learning approaches

In this study, six intelligent models, including LS-SVR ANFIS, and four ANN types, i.e., MLP, CFF, RBF, and GR neural networks, are applied to estimate the enhanced electrical efficiency of PV/T modules. It is necessary to determine the structural features of these ML techniques during the model development. Table 3 summarizes the most important structural features of the ML models and their investigated ranges in the current study. Numbers of the hidden neuron and learning algorithm must be determined for the MLP (Shafiq et al., 2021) and CFF (Amar, 2020) models. Moreover, the numbers of the hidden neuron and spread factor of the Gaussian transfer function in the hidden layer need to be determined appropriately for the RBF model (Daryayehsalameh et al., 2021). Since the numbers of the hidden neuron of the GR model dictated by the number of available training samples, the best spread value is needed to be found for this ANN type.

On the other hand, the most appropriate kernel type should be determined for the LS-SVR model (Hosseini and Vaferi, 2021). Fis type, number and center radius of clusters, and training algorithms must be chosen for the ANFIS-based model. Moreover, Table 3 shows that 720 CFFs, 840 MLPs, 350 RBFs, 100 GRs, 90 LS-SVRs, and 400 ANFISs are developed in this study. Indeed, we developed 2600 intelligent models and compared their accuracy to find the most precise AI-based model.

3.2. The best ML approach for estimation of electrical efficiency enhancement

It is worth noting that eighty percent of available databank (160 datasets) are used to train the intelligent models; the remaining completely unseen twenty percent (40 datasets) are used to check their generalization ability. The comparison study relies on the four statistical indexes, i.e., AARD%, MSE, RMSE, and regression coefficient (R^2) of the developed model for training, testing, and the whole of the database.

Table 4 reports the most accurate CFF selected among 720 models, MLP selected among 840 models, RBF selected among 350 models, GR selected among 100 models, LS-SVR selected

Table 3
The structural features of the employed ML methodologies and their investigated ranges in our study.

Computational method	Structural features	Investigated range	Numbers of the constructed model
Cascade feedforward	Numbers of the hidden neuron and learning algorithm	1–6 hidden neurons, 12 training algorithms	720
Multi-layer perceptron	Numbers of the hidden neuron and learning algorithm	1–7 hidden neurons, 12 training algorithms	840
Radial basis function	Numbers of the hidden neuron and spread factor	1–7 hidden neurons, spread of 1×10^{-5}–10	350
Generalized regression	Spread factor	Spread of 1×10^{-5}–10	100
LS-SVR	Kernel type	Three different kernels	90
ANFIS	FIS-type, numbers of the cluster, cluster center radius (radii), learning algorithm	Two FIS-types, 2–7 clusters, radii of 0.5 to 1, two training algorithms	400

Table 4
The most precise results obtained by different ML techniques.

Computational method	The best structural feature	Dataset	Uncertainty level			
			AARD%	MSE	RMSE	R^2
LS-SVR	Gaussian kernel	Training	15.74	3.367	1.84	0.9384
		Testing	20.54	3.614	1.90	0.9378
		Overall	16.70	3.417	1.85	0.9368
ANFIS	**Subtractive clustering FIS-type, radii = 0.667, Hybrid learning algorithm**	**Training**	**13.33**	**2.548**	**1.60**	**0.9502**
		Testing	**14.67**	**2.549**	**1.60**	**0.9668**
		Overall	**13.60**	**2.548**	**1.60**	**0.9534**
CFF	Six hidden neurons, Levenberg–Marquardt learning algorithm	Training	13.43	1.997	1.41	0.9634
		Testing	19.44	2.968	1.72	0.9394
		Overall	14.63	2.191	1.48	0.9597
MLP	Four hidden neurons, Levenberg–Marquardt learning algorithm	Training	15.51	2.365	1.54	0.9566
		Testing	18.38	3.764	1.94	0.9303
		Overall	16.08	2.645	1.63	0.9511
GR	160 hidden neurons, spread = 0.000001	Training	0.03	0.000	0.02	1.0000
		Testing	21.74	12.708	3.56	0.7308
		Overall	4.37	2.542	1.59	0.9535
RBF	Seven hidden neurons, spread = 3.4694	Training	21.59	5.455	2.34	0.9008
		Testing	18.86	3.369	1.84	0.9229
		Overall	21.04	5.037	2.24	0.9044

among 90 models, and ANFIS selected among 400 models. This table also presents the best characteristics for the considered artificial intelligent models. Up to now, we have chosen the six most accurate models among 2600 developed smart models. It is possible to find the most accurate smart model by comparing the accuracy of these six selected ML approaches. It is not hard to admit that the ANFIS model (highlighted by the bold font) shows better accuracy than the other developed models. This ANFIS model which is trained by the hybrid optimization algorithm has the subtractive clustering FIS-type and cluster center radius of 0.667.

The constructed ANFIS model predicts the training, testing, and overall datasets with the AARD% of 13.33, 14.67, and 13.6, respectively. The coefficients of determination of 0.9502, 0.9668, and 0.9534 are provided by the ANFIS model for training, testing, and overall datasets, respectively.

3.3. Assessment of machine learning model accuracy

This section aims to evaluate the performance of the developed ANFIS model to predict the enhancement in the electrical efficiency of the PV/T systems cooled by nanofluids through visual observation. A cross-plot of predicted versus actual electrical efficiency enhancements ($1\text{-}\eta_{nf}/\eta_{bf}$) for training and testing subdivisions is presented in Fig. 1. The dashed diagonal line shows the situation that predictions are precisely mapped on the actual data points. Since major parts of training and testing data points are concentrated around this diagonal line, it can be concluded that the model predictions have good agreement with the experimental datasets. It can also be seen that the proposed ANFIS model provides relatively high error for the prediction of only sixteen data points (four testing and twelve training).

The relative deviation between actual and predicted electrical efficiency enhancements is another index used to assess the ANFIS model accuracy. Fig. 2 illustrates the values of observed relative deviation for the collected experimental data points. This figure confirms that the designed ANFIS model cannot predict the small values of the dependent variable as accurately as the bigger ones. The arithmetic deviation between laboratory-measured and prediction values of enhancement in electrical efficiency (i.e., residual error) is also used to evaluate the model prediction reliability. Fig. 3 shows the histogram of the presented residual errors by the ANFIS model to estimate 200 experimental data points for the PV/T system electrical efficiency cooled by the nanofluids. It is easy to see that almost all data points are calculated with residual errors ranging from −4 to +4. On the other hand, only eight data points are predicted with residual errors out of this range. Moreover, an average value of −0.074% and a standard deviation of 1.599% are other justifications for the ANFIS's excellent performance.

3.4. Outlier detection

Since the proposed ANFIS model understands the PV/T system electrical behavior from the experimental datasets, it is necessary to check their validity. The experimental measurements may contain different levels of uncertainty (Skarbeli et al., 2021), fault

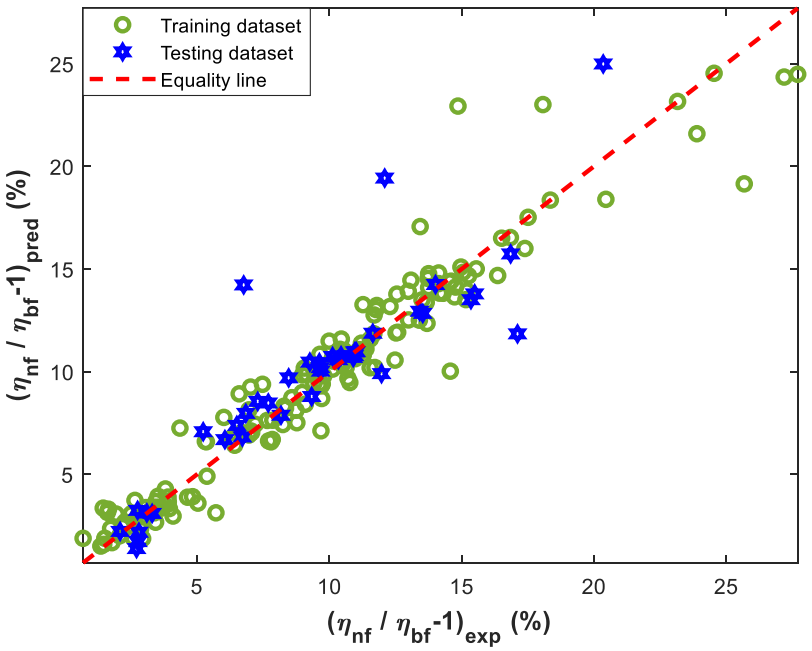


Fig. 1. Correlation between prediction and experimental data points of enhancement in electrical efficiency of PV/T systems by the nanofluids as coolant.

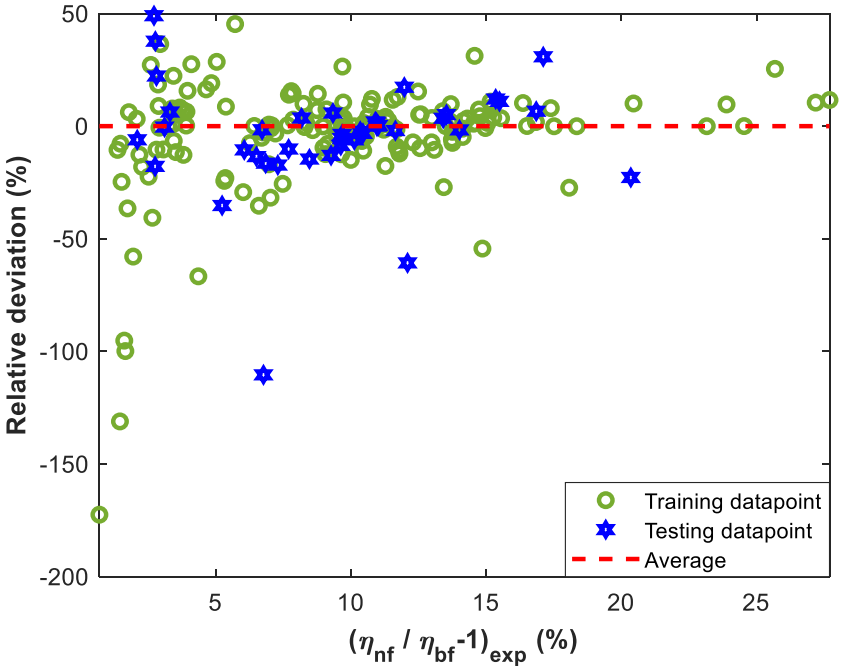


Fig. 2. Relative deviations between laboratory-measured and prediction values of enhanced electrical efficiency of PV/T systems.

(Fu et al., 2021), and outliers (Jiang et al., 2021; Zhou et al., 2021). This uncertainty in the experimental datasets poisons the understanding ability of the developed model for the considered phenomenon. The leverage method in William's domain is a reliable methodology to find valid and suspect data points (Daryayehsalameh et al., 2021; Pianko-Oprych et al., 2016). The valid term refers to those data that can be accurately predicted, and the suspect term includes those often estimated by low precision. In this graph, the warning leverage (H^*) is obtained from Eq. (10) (Moosavi et al., 2021):

$$H^* = 3(IV + 1)/N \tag{11}$$

In this equation, IV shows the number of independent variables (here is five), and N designates the number of available data points (here is 200). The typical value of three is often used for cut-off (representing a range of ± 3 standard deviations from the mean value of the datasets).

As Fig. 4 clearly shows, this method divides the domain into the valid and suspect regions. The outlier is referred those data that are placed in the suspect area. All other data points are valid information. From a modeling point of view, all data points are desired to be valid, and no outliers are attended. Fig. 4 confirmed that only ten out of two hundred collected datasets are outliers

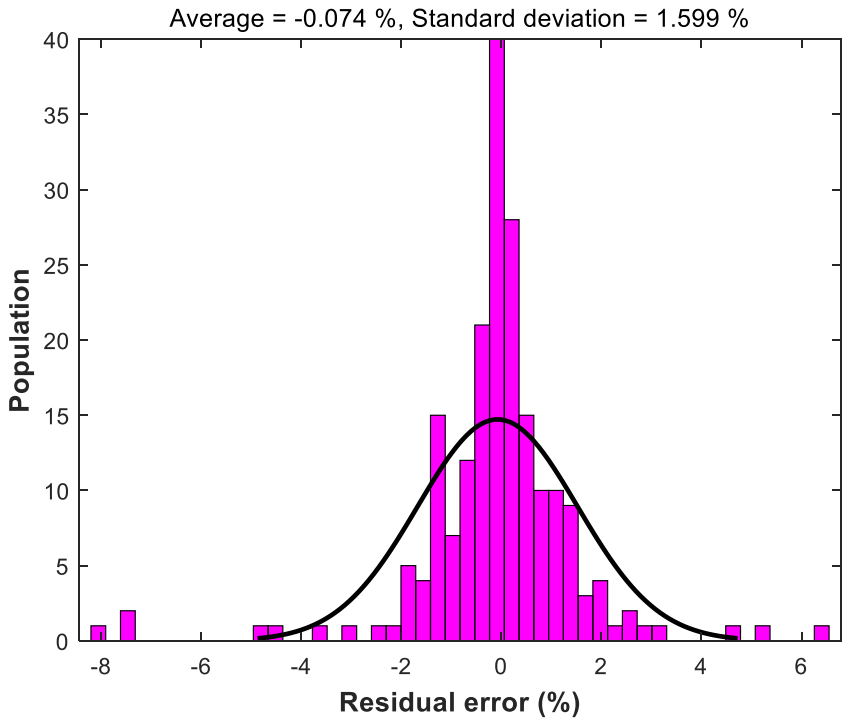


Fig. 3. Histogram of residual errors (actual minus predicted values) of enhancement in electrical efficiency of PV/T systems provided by the ANFIS model.

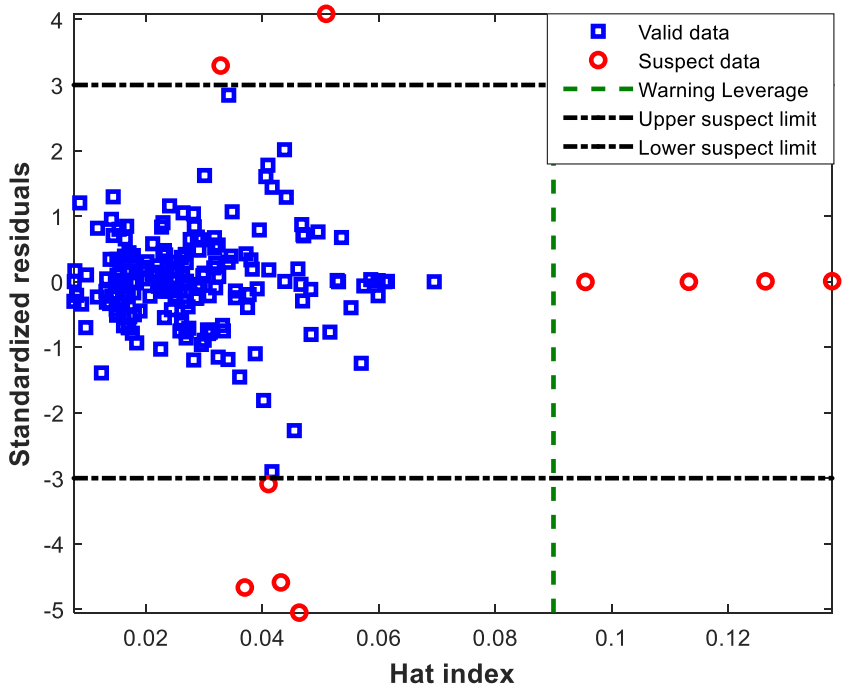


Fig. 4. Leverage analysis in William's graph for finding the outlier datasets.

(shown by the red circle symbol), and 95% are valid (depicted by the blue rectangular symbol). Six outliers and four reported measurements by the Ebaid et al. (2018) and Sardarabadi et al. (2014) are detected as outliers by the leverage method. The relative deviation between experimental measurements and ANFIS predictions for these outliers is in the range of 49%–175%. Generally, it can be said that the ANFIS model that was achieved its understanding from the valid database can be readily applied to predict the electrical efficiency of the PV/T systems cooled by the nanofluids in real situations.

3.5. The effect of influential variables on the enhanced electrical efficiency

3.5.1. Effect of coolant flow rate

This section investigates the effect of different independent variables on the enhanced electrical efficiency of the PV/T system cooled by nanofluids. The impact of radiation intensity and nanofluid flow rate (4 wt% of Ag nanoparticles in water) on enhancing the PV/T system electrical efficiency is presented in Fig. 5. Both laboratory measurements and their related predictions by

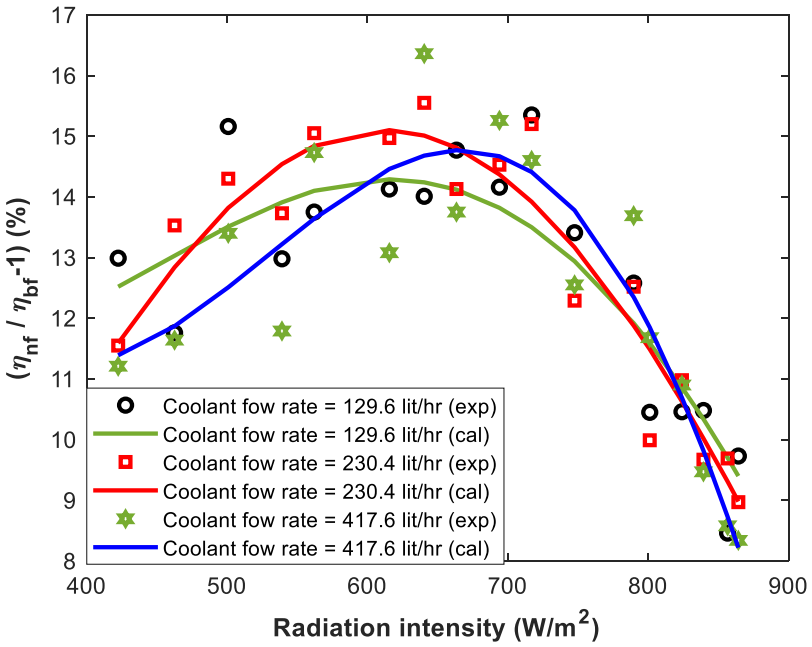


Fig. 5. Variation of enhanced electrical efficiency of the PV/T systems by radiation intensity and nanofluids flow rate (4 wt% of Ag nanoparticles with size of 20 nm).

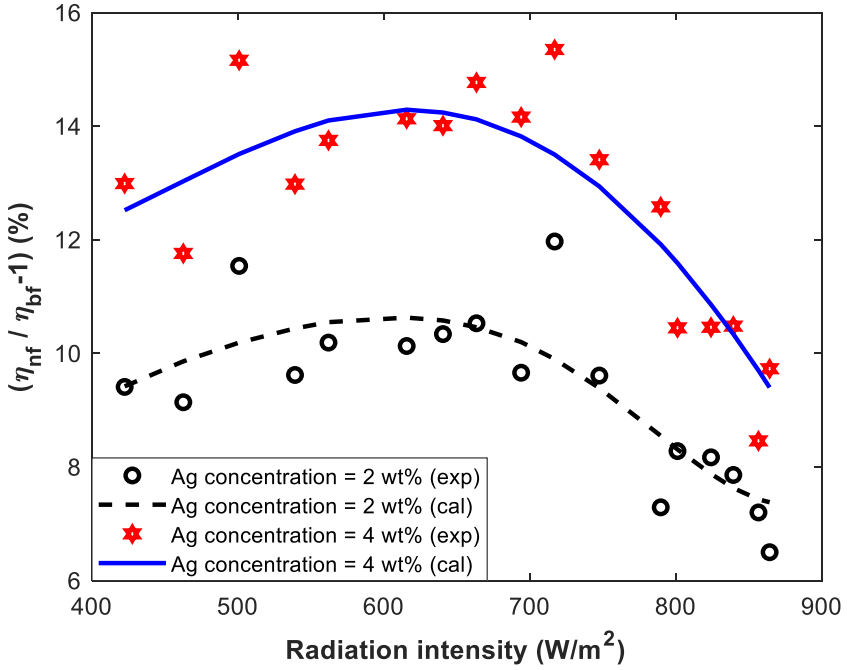


Fig. 6. Influence of radiation intensity and nanoparticle concentration on the enhanced electrical efficiency of the PV/T systems (129.6 l/h of Ag–water nanofluids).

the ANFIS models are plotted in this figure. First of all, a high level of scatterings can be simply observed in the experimental data points. The ANFIS model could efficiently handle this high level of scatterings and extract a reliable trend among them. The AARD of 5.86%, 4.28%, and 5.47% are observed for flow rates of 129.6, 230.4, and 417.6 l/h, respectively. It can be simply seen that it is almost impossible to present a general guideline for variation of the electrical efficiency by the flow rate of Ag–water nanofluids. On the other hand, increasing the radiation intensity up to a critical range (600–700 W/m^2) increases the PV/T electrical efficiency enhancement. Increasing the radiation intensity higher than the critical range reduces the enhanced electrical efficiency of the PV/T modules.

Furthermore, the proposed model allocates the maximum enhanced electrical efficiency to the highest flow rate for a higher radiation intensity. But the trend is inverse for a low radiation intensity (the experimental measurement errors, human mistake, model uncertainty, or/and some sorts of unknown phenomena may be responsible for this observation)

3.5.2. *Effect of nanoparticle concentration in water*

The effect of Ag nanoparticles concentration in the water on the PV/T electrical efficiency enhancement from the experimental and modeling perspectives is depicted in Fig. 6. This figure justifies that acceptable compliance exists between experimental data points and ANFIS predictions. Indeed, the AARD of

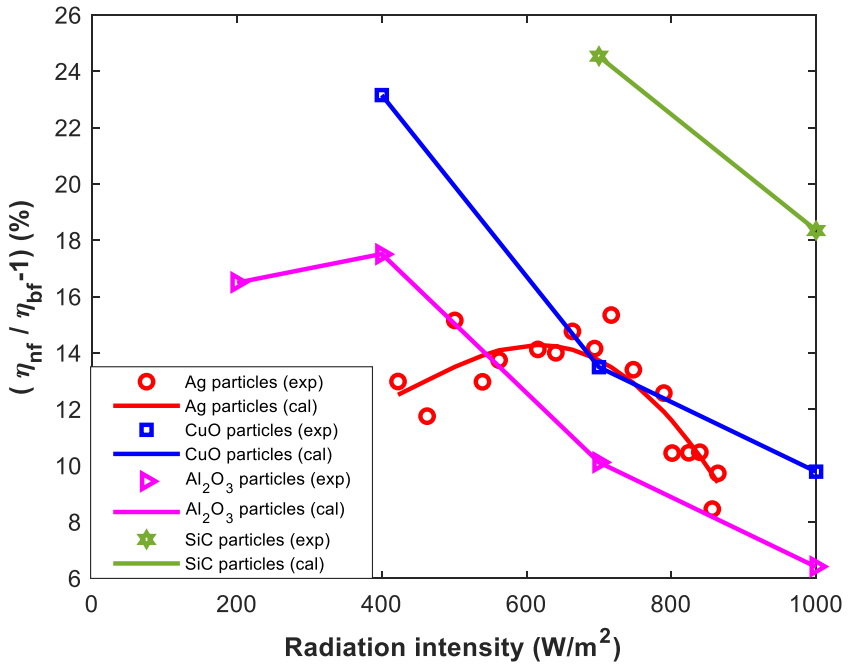


Fig. 7. Influence of nanoparticle type and radiation intensity on the enhanced electrical efficiency of the PV/T modules (4 wt% of nanoparticles in water).

6.21% and 5.86% are observed between actual and predicted Ag concentration values of 2 wt% and 4 wt%, respectively.

The electrical efficiency of the PV/T modules increases by increasing the Ag nanoparticle concentration. Improving the thermophysical characteristics of the nanofluids, including thermal conductivity (Khalifeh and Vaferi, 2019), convective (Vaferi et al., 2014) and pool boiling (Wood et al., 2019), heat transfer coefficient, viscosity (Gholami et al., 2018), and density by increasing the Ag concentration in water, are the main responsible for this finding.

3.5.3. Effect of nanoparticle type

The influences of the dispersion of 4 wt% of different nanoparticles on the PV/T electrical efficiency enhancement for experimental data and ANFIS predictions are illustrated in Fig. 7. This figure shows an acceptable agreement between laboratory measurements and the ANFIS predictions. Our proposed ANFIS model predicted the effect of SiC–water Ag–water, Al_2O_3 –water, and CuO–water nanofluids on the electrical performance of the PV/T system with AARD% of 5.86, 0.03, 0.06, and 0.03, respectively.

Generally, the SiC–water nanofluids produce the highest enhancement in the electrical efficiency of the PV/T modules. On the other hand, the alumina–water nanofluids result in the minimum improvement in the PV/T electrical efficiency.

Due to limitations in the experimental measurements, the nanoparticles' size is not equal, and the nanofluids may have different stability levels (Fig. 7). Therefore, these findings should be viewed as pure theoretical-based calculations.

3.6. Finding the optimum condition to maximize the PV/T electrical efficiency

This study develops an accurate model based on the ANFIS paradigm to predict the PV/T electrical efficiency as a function of five influential variables. This type of model can be readily coupled with an optimization algorithm to maximize the PV/T electrical efficiency. Although there are a lot of optimization algorithms (Edgar et al., 2001; Pianko-Oprych and Hosseini, 2017; Pourghebleh et al., 2021), this study uses the DE algorithm (Vaferi and Jahanmiri, 2011) to find the optimum point that maximizes the PV/T electrical efficiency. The DE algorithm and ANFIS model combination confirmed that three wt% of SiO_2 nanoparticles (diameter = 12.5 nm) at radiation intensity of 788.285 W/m^2 and 30 l/h of coolant is the optimum point that maximizes the electrical efficiency of the PV/T systems. In this optimum condition, the electrical efficiency enhancement of the PV/T system is 27.7%. It is worth noting that this optimum condition has been achieved inside the ranges of experimental measurements. The pure simulation results in Section 3.8 prove that a higher electrical efficiency enhancement can be obtained.

3.7. Validation of the ANFIS model by some external experimental samples

A databank containing 200 data samples has been used for intelligent models' development and testing. During the revision stage of this study, 63 new experimentally-measured data samples have been utilized for further validating the proposed ANFIS model (See Table 5). This new database includes 7 Ag–water, 30 Al_2O_3-water, 16 TiO_2-water, and 10 ZnO–water datasets. Table 5 shows that the proposed ANFIS model presents good predictions for this newly collected database. All experimental datasets have been estimated with excellent accuracy, excluding the PV/T system operating with the ZnO–water nanofluids predicted with the AARD = 34.77%. It should be mentioned that the constructed ANFIS model predicts the 63 newly collected datasets with the overall AARD = 15.21%.

3.8. Pure simulation results

This section presents pure simulation findings that have never been published before. Indeed, the reported results have been obtained for the ranges beyond the experimental domain reported in the literature (Aberoumand et al., 2018; Khanjari et al., 2016). Fig. 8 exhibits the effect of Ag size on the electrical efficiency enhancement of the PV/T collector. Although the literature presents some experimental measurements for the Ag nanoparticles with the diameter = 20 nm (dashed green line), there are no experimental and modeling results for the Ag nanoparticles with the

Table 5
Information of newly collected datasets from the literature.

Nanoparticle	Electrical efficiency increase (%)		AARD%	Reference
	Experimental range	ANFIS range		
Ag	1.51–2.72	1.57–2.53	6.28	Khanjari et al. (2016)
Al_2O_3	0.14–99.82	0.03–85.60	16.06	Khanjari et al. (2017, 2016) and Sangeetha et al. (2021)
TiO_2	63.16–99.83	70.31–95.21	5.31	Sangeetha et al. (2021)
ZnO	0.61–91.12	0.95–91.12	34.77	Hosseinzadeh et al. (2018)

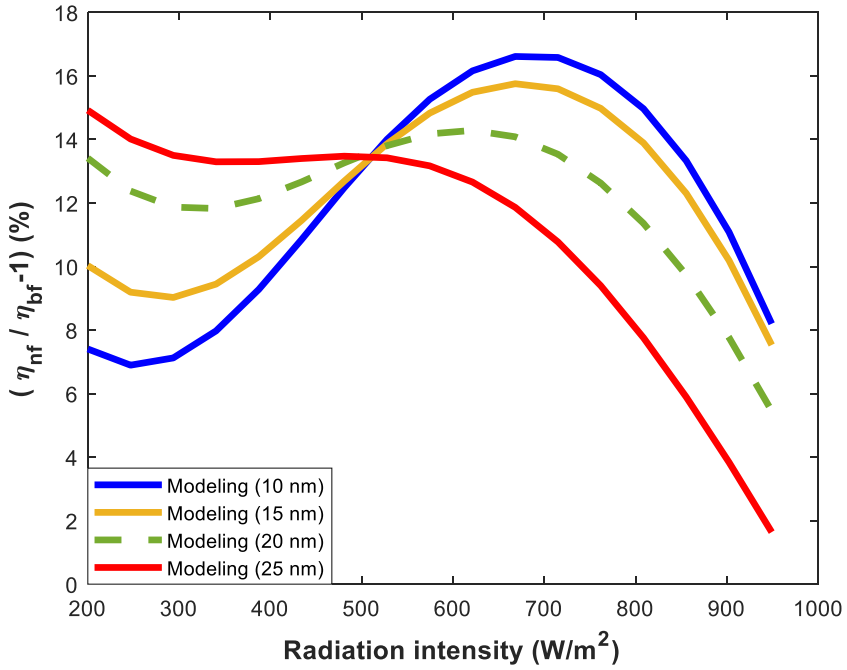


Fig. 8. Pure simulation profiles for investigating the effect of nanoparticle size on the enhancement of the electrical efficiency (Ag particle, 4 wt%, 129.6 l/h).

10, 15, and 25 nm. Fig. 9 shows a three-dimensional graph for variation of the electrical efficiency enhancement of the PV/T collector by the coolant flow and radiation intensity. Due to the expense and time required for doing experiments, the literature reported several limited measurements for this system (Al-Waeli et al., 2017; Khanjari et al., 2017, 2016; Sangeetha et al., 2021). The ANFIS model efficiently simulates the electrical efficiency enhancement overall ranges of interest.

3.9. Summary of the current study

The step-by-step procedures of this study are as follows:

1- Collecting experimental data from the literature (see Table 1)
2- Developing different artificial intelligent methods (see Table 3)
3- Finding the best AI-based model by combining trial-and-error process and statistical investigation (see Table 4)
4- Evaluating the reliability of the model using visual inspections (Section 3.3)
5- Employing the leverage method to monitor the valid/suspect experimental measurements (Section 3.4)
6- Using the best AI-based method to monitor the effect of independent factors on the enhanced electrical efficiency of the PV/T cells cools down by nanofluids (Section 3.5)
7- Coupling the proposed AI model and DE optimization algorithm to maximize the enhanced electrical efficiency of the PV/T cells operate with the nanofluids (Section 3.6)

4. Conclusion

Literature survey revealed that no accurate model is available to predict/optimize the PV/T systems' electrical efficiency cooled by nanofluids. Moreover, experimental studies reported ambiguous findings for the effects of nanofluid characteristics and operating conditions on the PV/T's electrical efficiency. The current study investigated the effect of radiation intensity, nanofluid flow rate, type, and concentration of nanoparticles in water on enhancing the PV/T electrical efficiency of the PV/T collectors. The accuracy of LS-SVR, four ANNs, and ANFIS models is compared using different statistical indices. To do so, 2600 models with different structural characteristics have been developed. Sensitivity analyses confirmed that the ANFIS with the subtractive clustering membership function and radius cluster of 0.667 is the best smart model for the considered matter. This model trained by the hybrid optimization algorithms using 160 experimental data provides excellent accuracy for simulation of the PV/T electrical performance. Indeed, it predicted 200 training and testing datasets with AARD = 13.6%, MSE = 2.548, RMSE = 1.6 and R^2 = 0.9534. Furthermore, a new external database containing 63 samples has successfully predicted with the AARD = 15.21%. The average value and standard deviation of −0.074% and 1.599% are observed for the residual deviation between actual measurements and ANFIS predictions. The leverage scenario justified that 95% of the collected experimental measurements (190 datasets) are valid. Both modeling and experimental observations confirmed that the PV/T electrical efficiency increases by increasing nanoparticle concentration in water. Moreover, the theoretical analysis showed that the SiC–water nanofluid is the best coolant

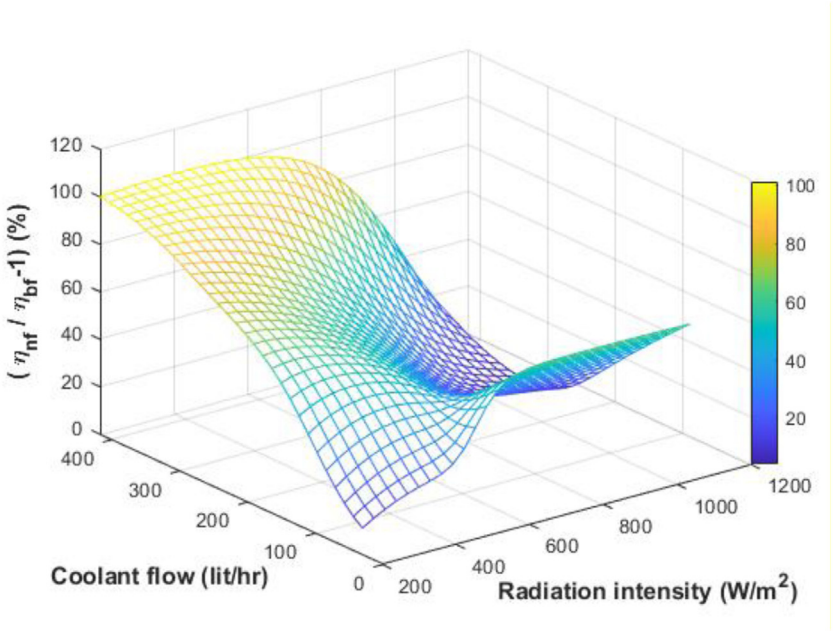


Fig. 9. Pure simulated three-dimensional graph reveals the effect of coolant flow rate and radiation intensity on the electrical efficiency enhancement (alumina nanoparticles, 50 nm, 2 wt%).

for the considered solar system. The effect of radiation intensity on the electrical efficiency of the PV/T systems is too hard to be generalized. Optimization by the differential evolution algorithm confirms that 30 l/h of water-silica nano-coolant (3 wt%, 12.5 nm) at radiation intensity of 788.285 W/m^2 maximizes the electrical efficiency of the PV/T system. In this optimum condition, the maximum enhancement in the PV/T electrical efficiency is 27.7%. Like all other data-driven methodologies, the constructed ANFIS model provides reliable results for interpolation, and its application for extrapolation purposes is not recommended.

CRediT authorship contribution statement

Yan Cao: Conceptualization, Methodology, Software. **Elham Kamrani:** Data curation, Writing - original draft. **Saeid Mirzaei:** Visualization, Investigation. **Amith Khandakar:** Formal Analysis, Methodology, Conceptualization. **Behzad Vaferi:** Investigation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The publication of this article was funded by the Qatar National Library.

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